

**KOÇ UNIVERSITY
SPRING 2026 SEMESTER
PHYS 201 – MECHANICS
COURSE INFORMATION**

Pre-requisites:	(PHYS. 101 and MATH. 204) or consent of the instructor
Instructor:	Özgür E. Müstecaplıoğlu Office: Sci 155, Phone: x1424 E-Mail: omustecap@ku.edu.tr
Office Hours:	By appointment.
TA:	Hamideh Gharib Masouleh (hmasouleh21@ku.edu.tr).
Textbook:	Stephen Thornton, Jerry Marion, “Classical Dynamics of Particles and Systems”, 5 th Ed., (Cengage Learning, 2003). Available as an eBook: https://www.caglayan.com/urundetay/755635/-#sthash.sdtNQTdN.HRYVPPKj.dpbs
Days & Times:	Feb. 09, 2026 – May. 22, 2026 Tue., Thu. 08:30 – 09:40 (Lectures) Fri. 13:00 – 14:10 (PS)
Access to Lectures:	Students can access the lectures in-person attendance in the classroom SNA 156 (PS location: SCI Z07).
Exams & Grading:	Midterm 30%, Final: 40%, Quizzes: 30%, Attendance [Bonus], Term paper [Bonus] A catalog grading system will be used: [90,100]: A, [85,89]: A-, [80,84]: B+, [75,79]: B, [70,74]: B-, [65,69]: C+, [55,64]: C, [50,54]: C-, [46,49]: D+, [40,45]: D, [0,39]: F. There will be no exams or assignments during the final week of the semester. All assignments and exams will be graded within ten days. The graded exams and assignments, along with solution keys, will be posted on the Learn Hub. Students who receive an F as their overall grade may participate in the remedial exam. Remedial exams replace the final exam grade in courses with a final exam. Students who did not attend the final will not be eligible for the remedial exam.
Term Paper:	Students may choose to write a term paper by proposing a topic to the instructor before the midterm date. The term paper must be completed as a team project consisting of three students. The paper must be submitted before the final week of classes. It will be considered a bonus assignment, with a maximum value of 10 points. An additional bonus of up to 5 points may be earned by delivering a short (10–15 minute) presentation of the term paper to the class during the finals week. To achieve the highest score, the term paper should include analytical derivations and/or numerical analysis, a critical review of relevant literature, and be well-written, logically structured, and properly cited, with sufficient diagrams to enhance clarity. The presentation grade will be based on time management, handling of the Q&A session, slide design and clarity, the quality of the summary, and the overall organization of the talk.

Students are encouraged to communicate with the instructor throughout the semester for feedback on their term paper, guidance on writing and revisions, and suggestions for improving their presentation.

Attendance:

Attendance will be recorded using a physical sign-in sheet circulated in class, on which students will be asked to write their names and provide a signature. Attendance may be counted as a bonus grade, with a maximum of 10 points. Each attended lecture corresponds to a bonus of $15/N$ where N is the total number of lectures held during the semester. Lectures will not be recorded. Attendance at both lectures and problem-solving (PS) sessions is strongly encouraged. In addition to enhancing students' problem-solving skills, PS sessions will include instruction on numerical methods using Python, Mathematica, and MATLAB for classical mechanics problems, which will be useful for completing numerical homework assignments.

As of the Fall 2025 semester, a minimum attendance rate of 70% is required for theoretical courses. Periods of absence due to illness are also included in the attendance record.

Quizzes

There will be no homework; however, problems from the textbook will be suggested at the end of each chapter. Quiz questions may be identical or closely related to these problems; therefore, solving the suggested problems is strongly recommended.

Exams

At the top of the very first page of the exam solution papers, the following honor code statement must be written and signed by the student:

"I pledge on my honor that I have not given or received any unauthorized assistance on this examination."

Authorized assistance is generally limited to the course textbook unless otherwise specified for a particular quiz or exam. Students may bring printed excerpts (not full copies) of the course e-textbook to quizzes and exams. However, these printed pages must not contain any additional notes. The TA and exam proctor will inspect the printed materials before allowing their use during the quiz or exam.

Make-up exams

Make-up exams can be administered for students if they have a university-approved excuse. The make-up exam dates are scheduled and announced by the Registrar's Office.

In case of medical excuses, students should submit their official health reports to Koç University Health Center for evaluation. If the KU Health Center approves the report, the student must contact the instructor to arrange a make-up exam.

In case of not being able to attend the final due to force majeure, a petition must be written to the relevant Faculty Executive Council before make-up exams.

Final exam

Final exam will be comprehensive, covering all the topics in the course. There are no special requirements to qualify for the final exam.

Course Subjects:

Newtonian mechanics (Ch 2), oscillations and phase space Ch 3.1-3.8), non-linear oscillations (Ch. 4), calculus of variations (Ch. 6), Lagrangian and Hamiltonian dynamics (Ch. 7.1-7.11), central-force motion (Ch. 8.1-8.8), dynamics of systems of particles (Ch. 9), motion in non-inertial frames (Ch. 10), dynamics of rigid bodies (Ch. 11), coupled oscillations (Ch. 12).

How to study this course?

Start by reading the course textbook, ideally before each lecture. If you come across any confusing points, make a note of them and ask about them during the lecture. Discuss any difficulties with the instructor, TA, or fellow students as soon as possible.

If you are still struggling with a concept, it can be helpful to see it explained in different ways. Try reading about the same topic in other books (see the list of recommended books in our library below). Eventually, a particular explanation or phrasing will help clarify your understanding. Additionally, online resources, lectures, and videos featuring experimental demonstrations can provide further insight.

You are strongly encouraged to discuss concepts, including homework problems, with your classmates. However, your homework and reports must reflect your own effort, with proper citations for any external sources used.

To solidify your understanding, practice by solving problems. If the textbook problems seem too challenging or the number of solved examples is insufficient, start with simpler problems from other books. Once you gain more experience, you can return to tackle the more difficult ones. If you can successfully solve textbook problems, you should have no difficulty achieving a high grade in this course.

Some recommended supplementary textbooks available in the library*:

- Classical mechanics, H. Goldstein.
- Mathematical Methods of Classical Mechanics, V. I. Arnol'd
- Mechanics, L. D. Landau & E. Lifshitz
- An Introduction to Mechanics, D. Kleppner & R. J. Kolenkow
- Calculus of Variations, R. Weinstock
- Classical Dynamics, S. T. Thornton, J. B. Marion
- The Feynman Lectures on Physics, Vol. I., R. Feynman
- Classical mechanics, R. D. Gregory
- Classical mechanics: a modern perspective, V. Barger & M. G. Olsson
- Classical mechanics: an undergraduate text, R. Douglas Gregory.
- Classical dynamics: a contemporary approach, Jorge V. José and Eugene J. Saletan.
- Berkeley physics course, Vol. 1

(*) Access and borrowing some titles from the library may be restricted. If you encounter problems in this regard, please contact the course instructor. There is a good collection of e-books in the library as well. There are many more titles in the library, as well as other excellent resources on the internet. Follow our course textbook but consult these additional resources when you struggle or get more curious about some concepts or problems. Some of these references are at a more advanced level, some are very conceptual, and some provide many more examples.

Academic Honesty: University policy regarding academic honesty will be fully enforced. Plagiarism is subject to disciplinary action. Please consult the university web page and the catalogue or your advisor for further information on this matter. Also see the attached document. By registering this course, it is assumed that the student is aware of these rules and regulations and fully accepts to obey them.

Koç University Statement on Academic Honesty with Emphasis on Plagiarism

Koç University expects all its students to perform course-related activities in accordance with the rules set forth in the Student Code of Conduct (<http://vpaa.ku.edu.tr/academic/student-code-of-conduct>). Actions considered as academic dishonesty at Koç University include but are not limited to cheating, plagiarism, collusion, and impersonating. This statement's goal is to draw attention to cheating and plagiarism related actions deemed unacceptable within the context of Student Code of Conduct: *All individual assignments must be completed by the student himself/herself, and all team assignments must be*

completed by the members of the team, without the aid of other individuals. If a team member does not contribute to the written documents or participate in the activities of the team, his/her name should not appear on the work submitted for evaluation.

Plagiarism is defined as ‘borrowing or using someone else’s written statements or ideas without giving written acknowledgement to the author’. Students are encouraged to conduct research beyond the course material, but they must not use any documents prepared by current or previous students, or notes prepared by instructors at Koç University or other universities without properly citing the source. Furthermore, students are expected to adhere to the Classroom Code of Conduct (<http://vpaa.ku.edu.tr/academic/classroom-code-of-conduct>) and to refrain from all forms of unacceptable behavior during lectures. Failure to adhere to expected behavior may result in disciplinary action.

There are two kinds of plagiarism: Intentional and accidental. Intentional plagiarism (Example: Using a classmate’s homework as one’s own because the student does not want to spend time working on that homework) is considered intellectual theft, and there is no need to emphasize the wrongfulness of this act. Accidental plagiarism, on the other hand, may be considered as a ‘more acceptable’ form of plagiarism by some students, which is certainly not how it is perceived by the University administration and faculty. The student is responsible from properly citing a source if he/she is making use of another person’s work. For an example on accidental plagiarism, please refer to the document titled “An Example on Accidental Plagiarism”.

If you are unsure whether the action you will take would be a violation of Koç University’s Student Code of Conduct, please consult with your instructor before taking that action.

An Example on Accidental Plagiarism: This example is taken from a document prepared by the City University of New York. The following text is taken from Elaine Tyler May's *'Myths and Realities of the American Family'*: *"Because women's wages often continue to reflect the fiction that men earn the family wage, single mothers rarely earn enough to support themselves and their children adequately. And because work is still organized around the assumption that mothers stay home with children, even though few mothers can afford to do so, child-care facilities in the United States remain woefully inadequate."*

Below, there is an excerpt from a student's homework, who made use of May's original text:
"As Elaine Tyler May points out, "women's wages often continue to reflect the fiction that men earn the family wage" (588). Thus many single mothers cannot support themselves and their children adequately. Furthermore, since work is based on the assumption that mothers stay home with children, facilities for day care in this country are still "woefully inadequate." (May 589)".

You may think that there is no plagiarism here since the student is citing the original author. However, this is an instance of accidental plagiarism. Although the student cites May and uses quotation marks occasionally, the rest of the sentences, more specifically the following section: "Thus many single mothers cannot support themselves and their children adequately. Furthermore, since work is based on the assumption that mothers stay home with children, facilities for day care in this country are still "woefully inadequate." (May 589)" almost exactly duplicates May's original language. So, in order to avoid plagiarism, the student either had to use quotation marks for the rest of the sentences as well, or he/she had to paraphrase May's ideas by using not only his/her own words, but his/her own original ideas as well. You should keep in mind that accidental plagiarism often occurs when the student does not really understand the original text but still tries to make use of it. Understanding the original text and understanding why you agree or disagree with the ideas proposed in that text is crucial both for avoiding plagiarism and for your intellectual development.

Reference(s):

Avoiding and Detecting Plagiarism: A Guide for Graduate Students and Faculty. The Graduate Center. City University of New York, 2012. Web. <http://www.gc.cuny.edu/CUNY_GC/media/CUNY-Graduate-Center/PDF/Publications/AvoidingPlagiarism.pdf>